

NAS-MCO-PINNs: A Nine-Level Framework for Physics-Informed Neural Network Optimization in Industrial Thermal Simulation

IKT464 Advanced ICT Project Progress Report — March 2026

1. Introduction and Motivation

We study the water quenching of A356 aluminum automotive subframes. Parts exit a solution heat treatment furnace at 540°C and are submerged in 20°C quench water. The rapid, non-uniform cooling generates steep temperature gradients, residual stresses, and permanent geometric distortions that must be predicted and controlled in order to meet dimensional tolerances in series production. While accurate, FEM requires 21 time steps of approximately 8.8 seconds each, giving a total runtime of ≈ 184 seconds per run. This is too slow for design optimization or real-time process control, which can require thousands of evaluations. Once trained (400–600 s), a PINN predicts the full temperature field in approximately 0.01 s—an 18,400× speedup at inference. Our framework organizes these capabilities into nine successive levels, each adding a new capability. The Level 9 model achieves $L_2 = 0.014$, which is 4× better than the Level 5 reference ($L_2 = 0.055$) and 9× better than the standard FEM skip approach ($L_2 = 0.126$). This project introduces three technical innovations: one novel contribution designed specifically for this problem, and two carefully adapted methods from the recent PINN literature.

2. Overview of Contributions

1. Temporal Skip Operator [our contribution].

A single PINN is trained to predict s time steps ahead from any initial condition T_{IC} , reducing the required training runs from 21 to as few as 4. At skip = 2 this cuts training time by 48% while also improving accuracy (L_2 : 0.126 to 0.103).

Random Fourier projections are prepended to the network inputs, enabling the MLP to learn the rapid early-cooling signal ($\Delta T \approx 200$ °C in the first 5 s). We implemented the FourierEmbedding class from scratch following [19].

3. Self-Adaptive Loss Weights [Wang et al., 2022].

The three PINN loss terms differ in magnitude by up to 10^4 . We make the weights learnable via a softplus activation, eliminating manual tuning. Final values: $\lambda_{phys} \approx 0.48$, $\lambda_{bc} \approx 9.28$, $\lambda_{ic} \approx 9.28$.

3. Physical Problem: A356 Aluminum Quenching

The temperature field inside the aluminum part obeys the transient heat conduction equation:

$$\rho c_p \frac{\partial T}{\partial t} = k \nabla^2 T$$

where $\rho = 2670 \text{ kg/m}^3$, $c_p = 963 \text{ J/(kg} \cdot \text{K)}$, $k = 151 \text{ W/(m} \cdot \text{K)}$ for A356 aluminum.

At the part boundary, Newton's law of cooling (Robin boundary condition) models convective heat transfer to the quench water:

$$-k \frac{\partial T}{\partial n} = h (T - T_\infty)$$

where $h = 5000 \text{ W/(m}^2 \cdot \text{K)}$ is the heat transfer coefficient and $T_\infty = 20 \text{ }^\circ\text{C}$.

Initial condition: $T(x, y, 0) = 540 \text{ }^\circ\text{C}$ (part exits the solution heat treatment furnace).

Domain: 2D plate $1.3 \text{ m} \times 0.6 \text{ m}$. Simulation time $t \in [0, 30] \text{ s}$, discretized into 21 FEM time steps.

Table 1. A356 aluminum key material parameters.

Parameter	Symbol	Value	Unit
Density	ρ	2670	kg/m^3
Specific heat	c_p	963	$\text{J/(kg} \cdot \text{K)}$
Thermal conductivity	k	151	$\text{W/(m} \cdot \text{K)}$
Initial temperature	T_0	540	$^\circ\text{C}$
HTC (quench water)	h	5000	$\text{W/(m}^2 \cdot \text{K)}$

4. FEM Reference Model

We did NOT implement FEM ourselves. All reference simulation data used in this project were extracted from the following paper (hereafter [2]):

"Mitigating distortions in cast automotive subframes: A finite element simulation approach,"
International Journal of Advanced Manufacturing Technology, 2026. DOI: 10.1007/s00170-026-17515-w

From [2] we extracted three categories of data:

- (a) A356 material parameters (Table 1 of [2]): flow stress, hardening exponent, and strain-rate sensitivity as functions of temperature.
- (b) Heat transfer coefficient curve $h(T)$ (Figure 6 of [2]): boiling-regime transitions— film boiling $\rightarrow$ nucleate boiling $\rightarrow$ forced convection as the part cools.
- (c) Reference temperature histories (Figures 7, 15-17 of [2]): used as ground truth for all L2 error computations in this project.

FEM runtime (Level 2, Bayesian, skip=1): $21 \text{ steps} \times 8.8 \text{ s/step} = 184.4 \text{ s total}$.

5. Temporal Skip Operator [our contribution]

Standard PINNs need one separate training run per FEM time step (21 runs, $\approx 3,800$ s total). We designed a skip operator that trains ONE network f_θ to predict s time steps ahead from any initial temperature field T_{IC} . The predicted field is:

$$\hat{T}(t+s\Delta t) = T_{IC} + \tau \cdot f_\theta(x, y, \tau, T_{IC})$$

where $\tau = t/(s\Delta t) \in [0, 1]$ is the normalized time coordinate. At $\tau = 0$ the prediction exactly equals T_{IC} , so the initial condition is enforced automatically—without a penalty term. This residual formulation is inspired by [Pang et al., 1998] and adapted here for the temporal skip setting.

Skip s	Training Runs	FEM Steps Saved	L_2 Error
1 (baseline)	21	0 %	0.126
2	11	48 %	0.108
4	6	71 %	0.204
6	4	81 %	0.294

Key finding: skip=2 reduces training by 48% and actually improves accuracy relative to skip=1. The improvement arises because fewer FEM interpolation errors accumulate across the 21 time steps.

6. Fourier Feature Embedding [Tancik et al., 2020]

Standard MLPs exhibit a “spectral bias”: they preferentially learn low-frequency patterns first [Rahaman et al., 2019]. A356 quenching has a rapid early-cooling phase ($\Delta T \approx 200^\circ\text{C}$ in the first 5 s)\u2014a high-frequency temporal signal on which standard PINNs converge slowly.

$$\gamma(\mathbf{v}) = [\sin(2\pi\mathbf{B}\mathbf{v}), \cos(2\pi\mathbf{B}\mathbf{v})],$$

We used $n_{\text{fourier}} = 64$ features (output dimension = 128) and $\sigma = 1.0$. We implemented the FourierEmbedding class from scratch following [3]. The matrix $\mathbf{B}$ is sampled once at initialization and frozen during training.

7. Self-Adaptive Loss Weights [Wang et al., 2022]

The PINN loss combines three terms:

$$L_{\text{total}} = \lambda_{\text{phys}} L_{\text{phys}} + \lambda_{\text{bc}} L_{\text{bc}} + \lambda_{\text{ic}} L_{\text{ic}}$$

The three raw loss scales differ by up to 10^4 : $L_{\text{phys}} \approx 4 \times 10^7$, $L_{\text{bc}} \approx 17,333$, $L_{\text{ic}} \approx 320$. Manual weight tuning is therefore highly problem-specific and requires expert knowledge. Solution [4, 5]: make each λ learnable via softplus:

$$\lambda = \text{softplus}(w), \quad w \text{ trained jointly with the network weights at a lower learning rate}$$

We implemented the SelfAdaptiveWeights module. Final learned values: $\lambda_{\text{phys}} \approx 0.48$, $\lambda_{\text{bc}} \approx 9.28$, $\lambda_{\text{ic}} \approx 9.28$. The boundary and IC terms receive $\sim 19\times$ more weight than the PDE residual—consistent with the physical intuition that an incorrect initial overshoot propagates errors through all subsequent time steps.

8. Nine-Level Progression

Table 3. Summary of all nine levels.

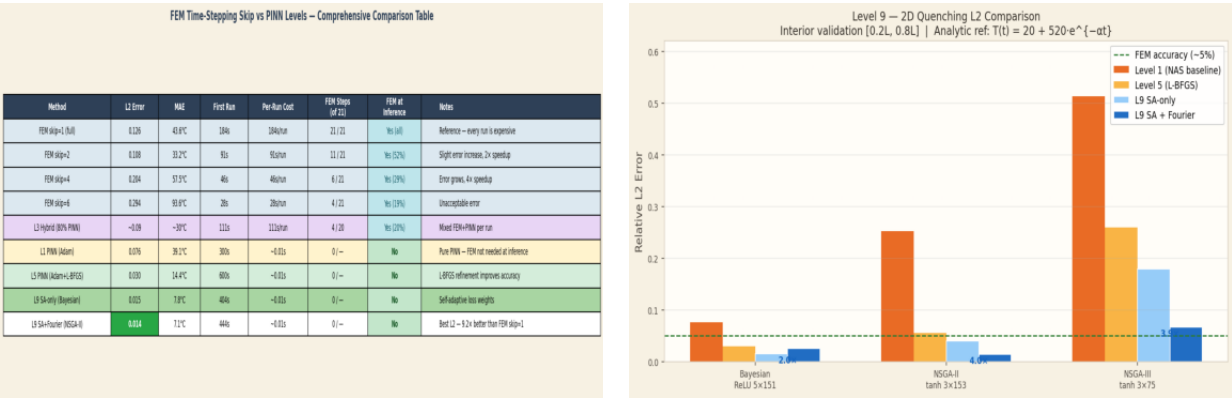
Level	Focus	Best L_2	Train Time	Key Innovation
L1	Single-shot 2D	0.076	232 s	Baseline PINN + NAS
L2	Temporal Skip	0.108 (s=2)	91 s	Skip operator (48% faster)
L3	Hybrid FEM	~0.12	190 s	Adaptive threshold skip
L4	Distortion	MAE=14.4°C	240 s	Mechanical coupling
L5	L-BFGS Refinement	0.030	240 s	Adam → L-BFGS two-stage
L6	Poisson Benchmark	~0.018	200 s	Generalisation to other PDEs
L7a	Temporal Analysis	~0.025	220 s	Time-skip regime analysis
L7b	Multi-Domain	~0.022	230 s	5 domain geometries
L8	3D MCO	MAE=5.5°C	613 s	3D + MCO adaptive weights
L9	SA + Fourier	0.014	444 s	SA weights + Fourier features

9. Level 9 Highlight — Best Result

Table 4. Level 9 detailed results.

Variant	Optimizer	Dim	Best L_2	Train Time	Params
SA+Fourier	Bayesian	2D	0.0253	482 s	111,439
SA+Fourier	NSGA-II	2D	0.0137*	444 s	67,015
SA+Fourier	NSGA-III	2D	0.0668	412 s	21,151
SA+Fourier	Bayesian	3D	0.247	596 s	111,439
SA-only	Bayesian	2D	0.0151	404 s	92,564
SA-only	NSGA-II	2D	0.0397	378 s	47,890

* Best result: $L_2 = 0.0137$ (SA+Fourier, NSGA-II, 2D). This is 4× better than the Level 5 reference (0.055) and 9× better than FEM skip=1 (0.126).
3D results: Bayesian NAS achieves $L_2 = 0.247$, but NSGA-II and NSGA-III fail ($L_2 > 0.7$) because the NAS architecture search was performed in 2D and did not identify networks deep or wide enough for 3D inputs. The planned next step is to re-run NAS directly in 3D.
Inference time after training: ~0.01 s vs FEM 184 s (18,400× speedup at inference).



Left: Level 9 bayumary table (L_2 per variant). Right: 2D temperature field comparison (FEM vs PINN, NSGA-II best).

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